## Binomial $\longrightarrow$ Normal

(for large $n$ and $npq$ not too small)

$P(X=k) = \binom{n}{k} p^k q^{n-k}$

$X \sim \text{binomial}(n, p)$

# successes in $n$ indep. Bernulli trails

$X = B_1 + B_2 + \dots + B_n$ where $B_i \sim \text{Bernulli}(p)$

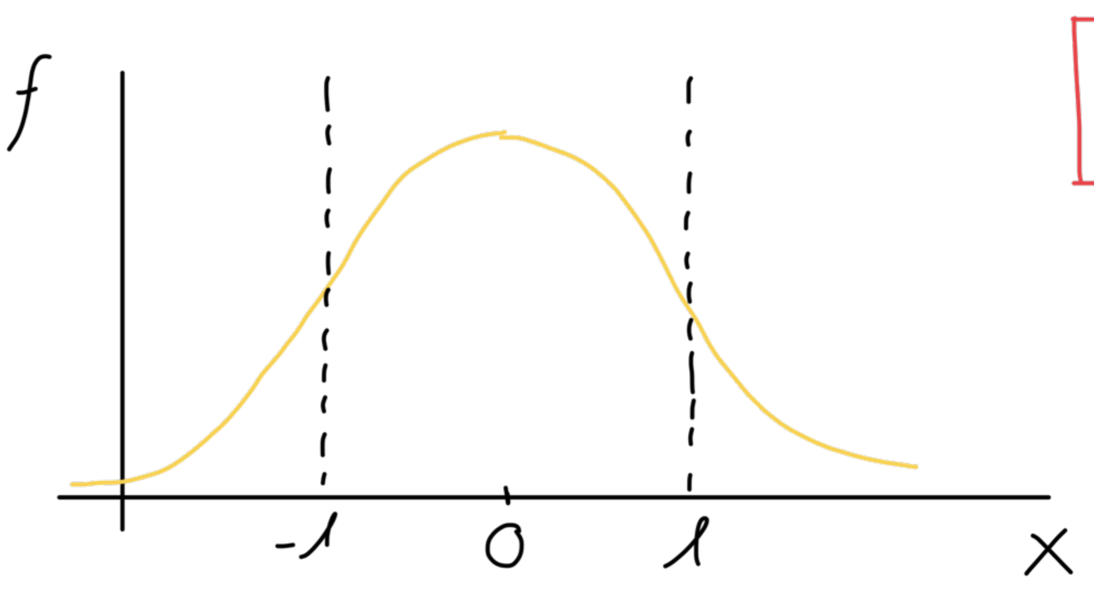
$$X \sim \text{normal}(\underbrace{np}_{\mu}, \underbrace{npq}_{\sim \sigma^2})$$

$$P(X=x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

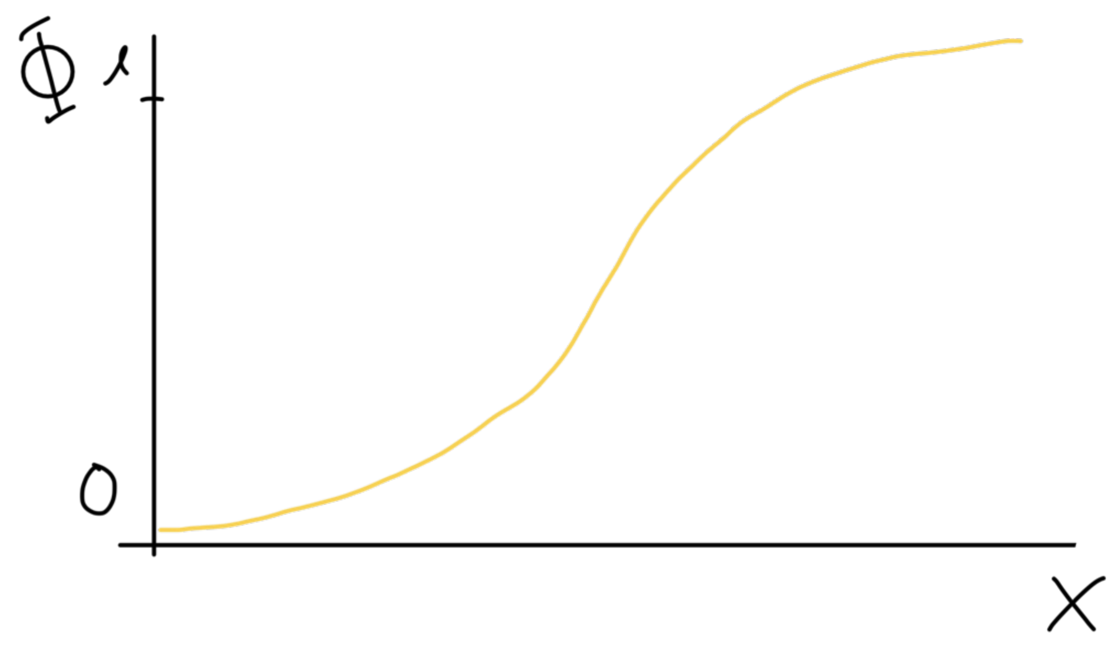
Easier to compute with $N \sim (\mu, \sigma^2)$!

Standard unit Normal $(\mu = 0, \sigma = 1)$

PDF




CDF




$$P(X \le x) = \int_{-\infty}^{x} f(x)\,dx$$

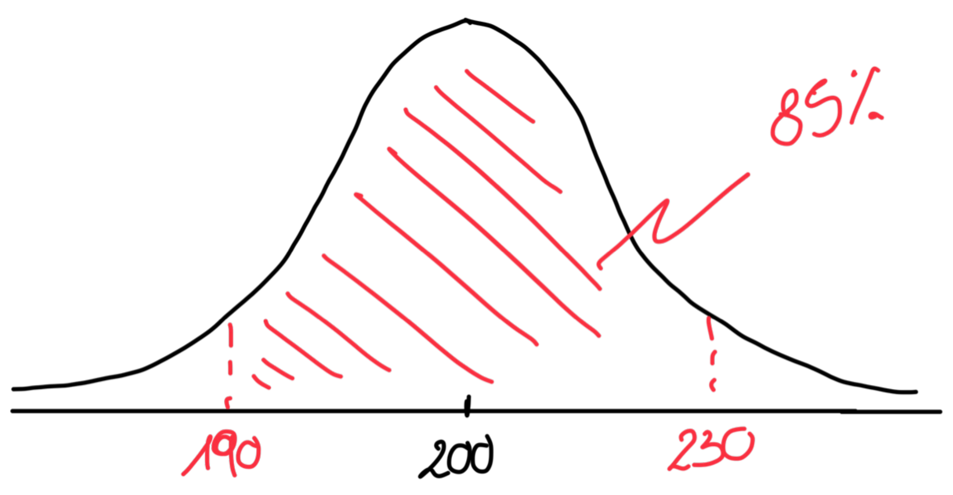
$$P(-1 \le X \le 1) = \int_{-1}^{1} f(x)\,dx = \Phi(1) - \Phi(-1)$$

Ex:

Flip a coin 400 times ; # heads $= X \sim \text{binomial}(400; \tfrac{1}{2})$

$P(190 \le X \le 230) = P(x=190) + P(x=191) + \dots + P(x=230)$

$X \sim \text{normal}(200, 100)$




## Binomial $\longrightarrow$ Poisson

Small p

Let $\lambda = np$

$X \sim \text{binomial}(n, p)$

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k} =$$

$$= \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} =$$

$$= \frac{\lambda^k}{k!} \overbrace{\frac{n(n-1)(n-2)\dots(n-k+1)}{n \cdot n \cdot n \dots n}}^{\to 1} \left(1 - \frac{\lambda}{n}\right)^n \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-k}}_{\to 1} =$$

as $n \to \infty$ $\quad = \frac{\lambda^k}{k!} \cdot e^{-\lambda}$

$$\boxed{P(X=k) = \frac{\lambda^k}{k!} \cdot e^{-\lambda}}$$

Ex: Prob. of one light going out $p = 0.002$

$n = 10'000$ $\quad \lambda = 20$

$$\text{Prob}\left(\begin{array}{c}\text{exactly } k \\ \text{lights going} \\ \text{out}\end{array}\right) \sim \text{Pois}(20)$$